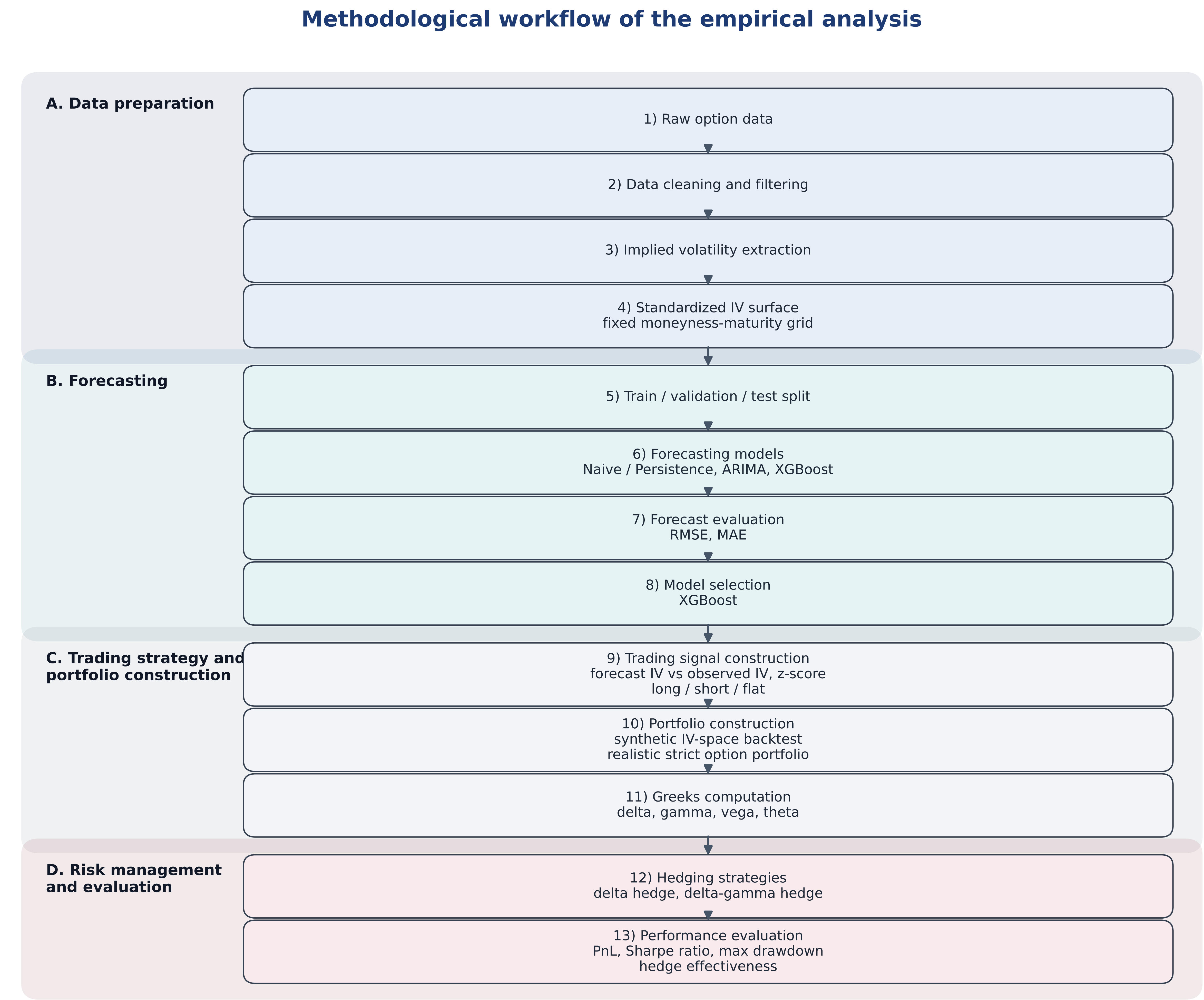
Methodological workflow of the empirical analysis
A. Data preparation
1) Raw option data
2) Data cleaning and filtering
3) Implied volatility extraction
4) Standardized IV surface
fixed moneyness-maturity grid
B. Forecasting
5) Train / validation / test split
6) Forecasting models
Naive / Persistence, ARIMA, XGBoost
7) Forecast evaluation
RMSE, MAE
8) Model selection
XGBoost
C. Trading strategy and portfolio construction
9) Trading signal construction
forecast IV vs observed IV, z-score
long / short / flat
10) Portfolio construction
synthetic IV-space backtest
realistic strict option portfolio
11) Greeks computation
delta, gamma, vega, theta
D. Risk management and evaluation
12) Hedging strategies
delta hedge, delta-gamma hedge
13) Performance evaluation
PnL, Sharpe ratio, max drawdown
hedge effectiveness